

Homework 4

Multiple Linear Regression

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Data and Description

Measuring body fat is complex. One method involves submerging the body in water and measuring displacement. A simpler, more accessible method would be preferable. To develop such a method, researchers collected data from 252 men, recording age (years), weight (pounds), height (inches), and three body circumference measurements (neck, chest, and abdomen, all in centimeters). Each participant's body fat percentage (variable "brozek") was accurately estimated using underwater weighing. The goal is to create a predictive model using these basic variables, eliminating the need for water displacement measurements.

The dataset, `BodyFat.txt`, is available on Canvas. Download `BodyFat.txt` and place it in the same folder as this `.qmd` file.

0. Replace the text "< PUT YOUR NAME HERE >" (above next to "author:") with your full name.

1. Read in the data set, and call the data frame "bodyfat". Display a summary of the data. Remove the "row" column (which contains row numbers) from the data set.

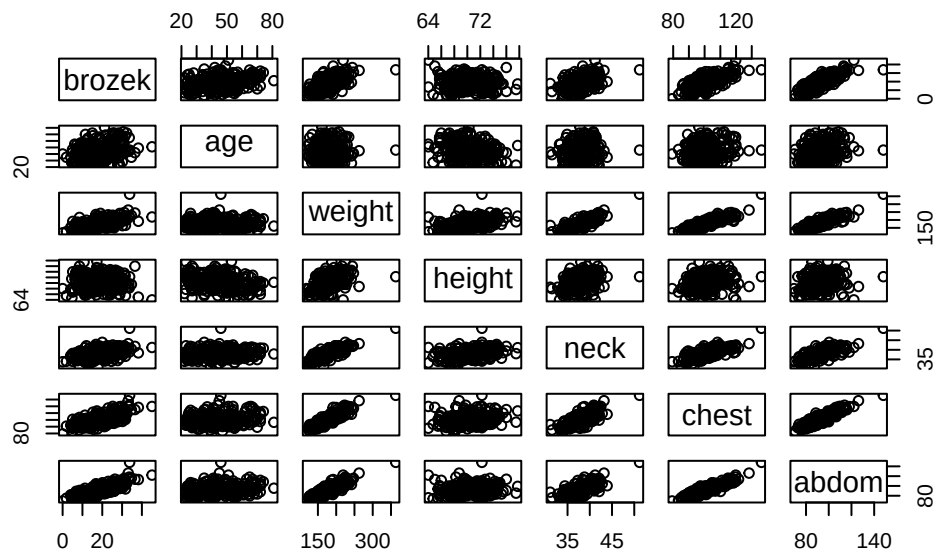
```
bodyfat <- read.table("BodyFat.txt", header = TRUE)
bodyfat <- bodyfat %>% select(-row)
summary(bodyfat)
```

brozek	age	weight	height
Min. : 0.00	Min. :22.00	Min. :118.5	Min. :64.00
1st Qu.:12.80	1st Qu.:35.50	1st Qu.:158.8	1st Qu.:68.38
Median :19.00	Median :43.00	Median :176.2	Median :70.00
Mean :18.89	Mean :44.86	Mean :178.8	Mean :70.32
3rd Qu.:24.55	3rd Qu.:54.00	3rd Qu.:196.9	3rd Qu.:72.25

Max. :45.10	Max. :81.00	Max. :363.1	Max. :77.75
neck	chest	abdom	
Min. :31.10	Min. : 79.3	Min. : 69.40	
1st Qu.:36.40	1st Qu.: 94.3	1st Qu.: 84.55	
Median :38.00	Median : 99.6	Median : 90.90	
Mean :37.98	Mean :100.8	Mean : 92.49	
3rd Qu.:39.40	3rd Qu.:105.3	3rd Qu.: 99.20	
Max. :51.20	Max. :136.2	Max. :148.10	

2. Display a scatterplot matrix of the data.

```
plot(bodyfat)
```



3. Based on the scatterplot matrix, briefly explain which variables you think will be helpful for predicting brozek and which variables you think will *not* be helpful at predicting brozek. Explain how the scatterplot helped determine your answers.

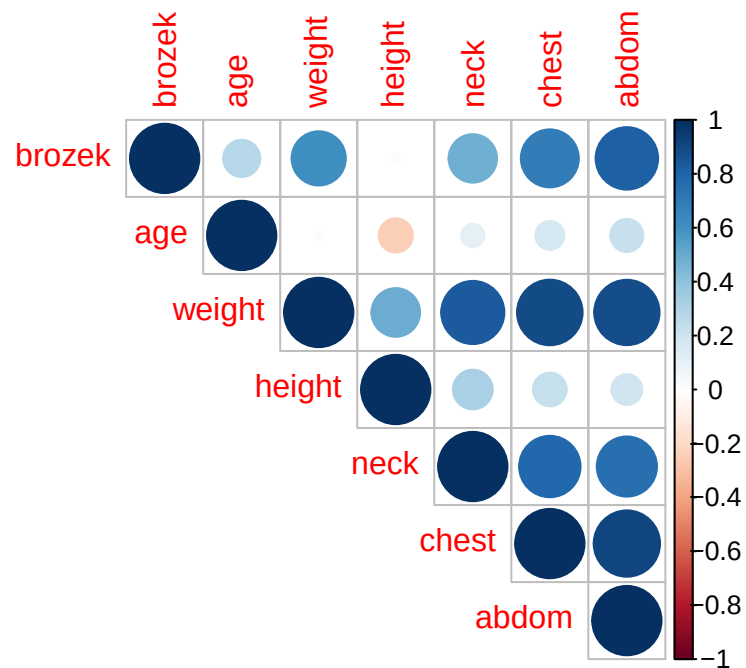
Looking at the scatterplots, I think that weight, neck, chest, and abdomen are going to be the most helpful for predicting brozek, and that age and height will be less helpful. I think so because those first four show obvious positive trends when plotted against brozek, while the others are more scattered or flat.

4. Create and print a correlation matrix (numeric or color- and shape-coded).

```
round(cor(bodyfat), 2)
```

	brozek	age	weight	height	neck	chest	abdom
brozek	1.00	0.29	0.61	-0.02	0.49	0.70	0.81
age	0.29	1.00	-0.01	-0.24	0.11	0.17	0.23
weight	0.61	-0.01	1.00	0.49	0.83	0.89	0.89
height	-0.02	-0.24	0.49	1.00	0.33	0.24	0.20
neck	0.49	0.11	0.83	0.33	1.00	0.78	0.75
chest	0.70	0.17	0.89	0.24	0.78	1.00	0.92
abdom	0.81	0.23	0.89	0.20	0.75	0.92	1.00

```
corrplot(cor(bodyfat), type = "upper")
```



5. Based on the scatterplot matrix and the correlation matrix, are there any pairs of variables that you suspect will cause a problem in terms of multicollinearity? If so, which ones?

There are at least a few. The most worrisome would probably be weight compared with neck, chest, or abdomen, neck with chest or abdomen and chest with abdomen.

6. Fit a multiple linear regression model to the data (no transformations). Display a summary of the results.

```
linear_model <- lm(data = bodyfat, formula = brozek ~ .)
summary(linear_model)
```

Call:

```
lm(formula = brozek ~ ., data = bodyfat)
```

Residuals:

Min	1Q	Median	3Q	Max
-11.5811	-3.0358	0.0668	2.8879	10.2828

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.010e+01	1.413e+01	-1.423	0.1561
age	5.010e-03	2.540e-02	0.197	0.8438
weight	-8.733e-02	3.879e-02	-2.251	0.0253 *
height	-1.400e-01	1.520e-01	-0.921	0.3579
neck	-4.421e-01	2.019e-01	-2.189	0.0295 *
chest	4.844e-04	9.107e-02	0.005	0.9958
abdom	8.754e-01	7.924e-02	11.048	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.124 on 244 degrees of freedom

Multiple R-squared: 0.722, Adjusted R-squared: 0.7151

F-statistic: 105.6 on 6 and 244 DF, p-value: < 2.2e-16

7. Briefly comment on the “significance” of the variables: were you surprised by the results? Are there any variables that are significant that you think shouldn’t be? Are there any variables that are not significant that you think should be?

It makes sense that weight and abdomen would be significant, but it doesn’t seem like neck should be over some of the other variables measured. I also feel like chest should be significant as well.

8. Briefly comment on the sign (+/-) of the coefficients for the variables. Are there any variables where the sign is the opposite of what you expected?